

A finite recurrent optimal-design workflow for RLS rTMS research

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Nature-style research preprint | Synthetic simulation | Not peer reviewed

ABSTRACT

This preprint documents a finite, synthetic modeling workflow for a restless legs syndrome (RLS) rTMS research dashboard. A five-target cortical allocation is propagated through a bounded recurrent update, a finite experimental-design grid, a normalized boundary-field visualization, simulated response-score draws, and a hypothetical market scenario. The implementation is intended for software visualization and reproducibility only. It does not establish efficacy, estimate cure probability, prescribe stimulation, or support a clinical decision.

1. Finite model

The model represents five predefined target allocations spanning medial frontal, frontal, parietal, temporal, and occipital cortical regions. These labels are design variables for a visualization experiment, not a proposal to stimulate all cortical lobes clinically. Each target receives a bounded weight and evolves through a finite recurrence over the study horizon.

$$a_{\text{ell},k+1} = \min(1, 0.63 a_{\text{ell},k} + w_{\text{ell}} [1 - \exp(-0.42k)]), \text{ ell} = 1, \dots, 5$$

Equation 1 | Finite recurrent state update for the lobe allocation visualization.

$$D(f, I) = \exp(-((f-10)/7)^2 - ((I-100)/18)^2); (f^*, I^*) = \operatorname{argmax}_{(f, I)} D(f, I)$$

Equation 2 | Bounded 4 by 4 experimental-design score; it is not a treatment optimizer.

$$F_{\text{pq}} = \sum_{\text{ell}=1}^5 w_{\text{ell}} \exp(-((x_{\text{p}}-c_{\text{x},\text{ell}})^2 + (y_{\text{q}}-c_{\text{y},\text{ell}})^2)/0.13)$$

Equation 3 | Finite normalized multi-source boundary-field visualization.

$$R_k = 100 [B - S_k] / B, k = 0, \dots, K$$

Equation 4 | Synthetic recovery percentage from baseline index B.

$$\text{Rev}_y = N_{\text{courses}} P_{\text{course}} g_y; \text{NPV} = -C_0 + \sum_{y=1}^5 (\text{Rev}_y - 0) / (1+r)^y$$

Equation 5 | Finite five-year revenue and discounted-cash-flow scenario.

2. Characteristics and figures

Characteristic	Finite-model output	Scope
Cortical targets	5	Predefined research allocation
Experimental design	10 Hz / 100%	Finite-grid maximum
Response-score likelihood	83.1%	Synthetic draw threshold $\geq 50\%$
Horizon recovery	54.2%	Synthetic model output
Market scenario	\$5,829,384 / 5 years	Hypothetical clinic model

Table 1 | Characteristics generated during this run. Each result is restricted to the model's finite assumptions.

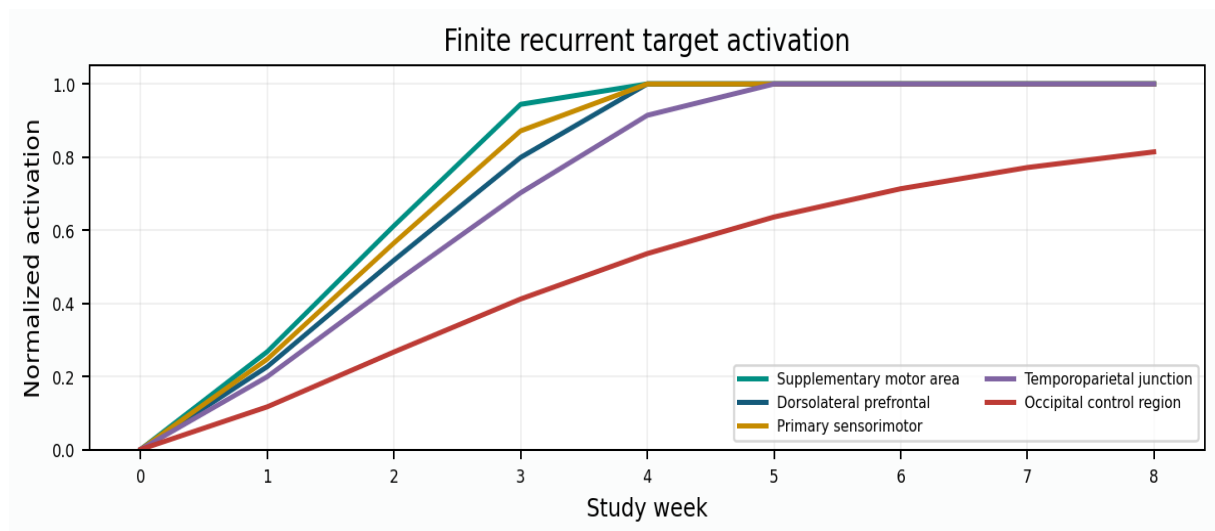


Figure 1 | Recurrent activations for the predefined target allocation. Curves are synthetic state variables.

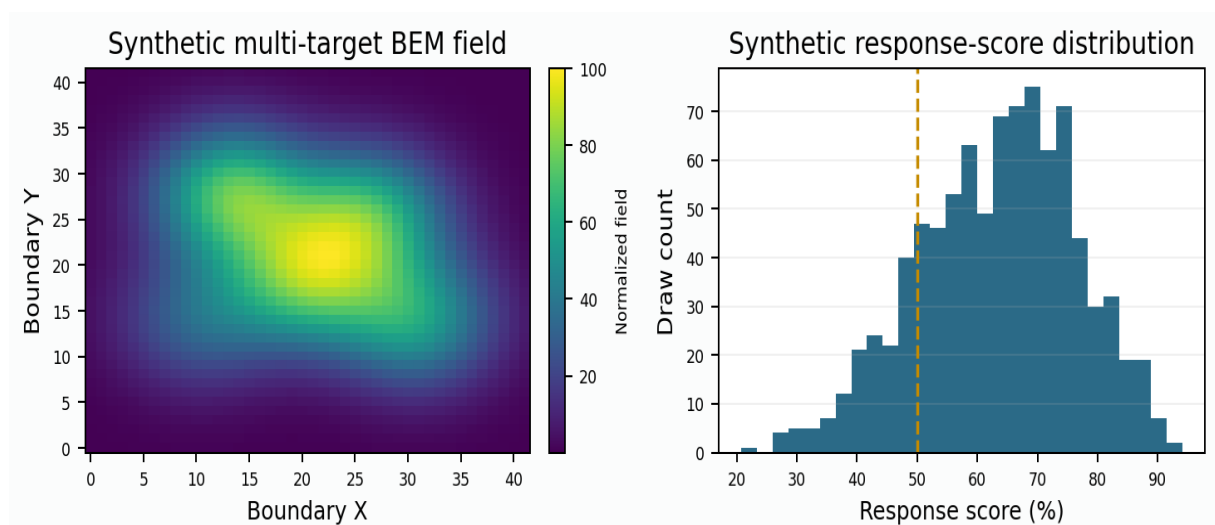


Figure 2 | Normalized BEM-like target-field visualization and synthetic response-score distribution. Neither panel is in-vivo dosimetry or an outcome forecast.

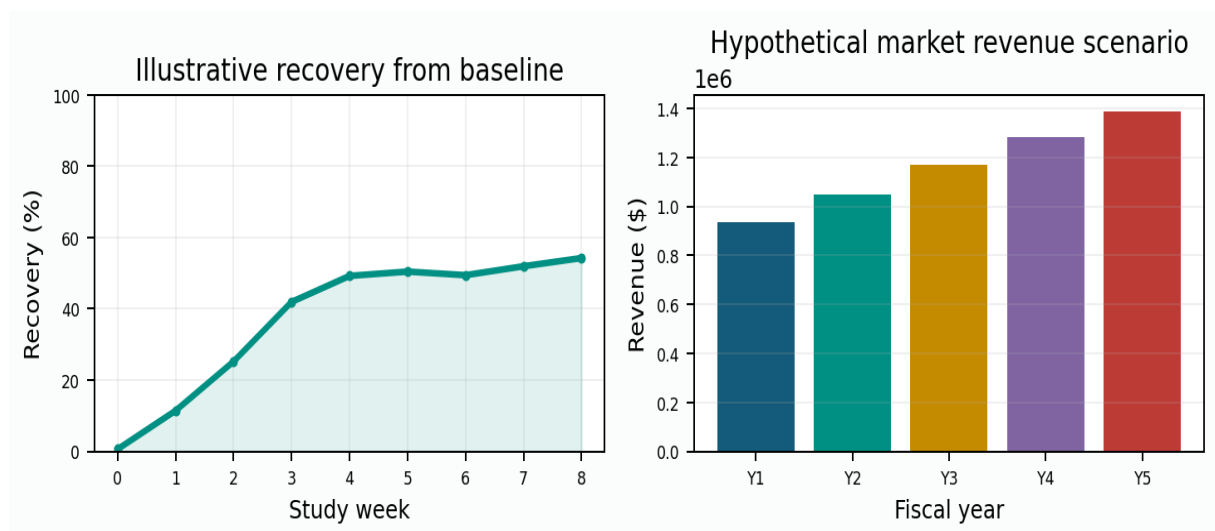


Figure 3 | Illustrative model recovery and hypothetical five-year clinic-market revenue scenario.

3. Equipment and economic assumptions

The dashboard models a clinic with figure-8 TMS systems, neuronavigation, EMG/accelerometry, cooling, and safety monitoring. Quantities are operational placeholders. Revenue is computed from modeled annual course capacity and modeled course price, while costs are represented through explicit capital and operating assumptions. These financial outputs are scenario calculations rather than market forecasts.

4. Limitations

No participant data, patient anatomy, stimulator waveform calibration, clinical comparator, safety monitoring data, regulatory clearance, or external outcome validation were used. The response distribution is sampled from a fixed beta distribution and must not be interpreted as a probability of recovery or cure. Any clinical study would require appropriate ethical review, registered protocol, validated endpoints, and qualified clinical oversight.

Code availability

The local workspace contains the RLS simulation module and this PDF generator. Running the generator recalculates the finite values and embeds the resulting figures.

Research-only computational preprint. Not medical advice, not a clinical protocol, and not evidence of efficacy.